

Package ‘bulknul’

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Title Assessing Whether a Bulk-Null Result Can Distinguish Cell-State
Dilution from Genuine Absence

Version 0.1.0

Description Diagnostics for bulk transcriptomic null results in studies that
also have single-cell data. Computes a Dilution Discordance Score, the
posterior probability that a non-significant bulk effect reflects dilution
of a small subcluster rather than genuine absence, together with a
Diagnosability Index quantifying whether the study design has power to make
that distinction at all. Includes an applicability gate that refuses
bulk-significant inputs.

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bulknull

One-Call Wrapper for Bulk-Null Dilution Diagnostics

Description

Unified interface that runs the applicability gate, computes the Dilution Discordance Score (DDS), and the Diagnosability Index (DI) in a single call. Returns a formatted S3 object summarizing the complete analysis.

Usage

```
bulknull(
  bulk_beta,
  bulk_se,
  bulk_fdr,
  d_sc = NULL,
  sc_zscore = NULL,
  cluster_fraction,
  condition_fraction,
  n_cluster = NULL,
  n_eff_basis = c("cell", "donor"),
  n_donors = NULL,
  null_fdr_threshold = 0.05,
  on_violation = c("error", "warn", "pass"),
  alpha = 0.05,
  sided = c("one", "two")
)
```

Arguments

bulk_beta	numeric; effect size estimate from bulk analysis
bulk_se	numeric; standard error of bulk_beta
bulk_fdr	numeric; FDR (or p-value) from bulk test
d_sc	numeric; standardized effect size in single-cell cluster (primary input)
sc_zscore	numeric; z-score of signal in target cluster (backward compatibility)
cluster_fraction	numeric; fraction of total cells in target cluster
condition_fraction	numeric; fraction of cells in case condition
n_cluster	numeric; number of cells in target cluster (required if sc_zscore supplied)
n_eff_basis	character; basis for sc_zscore conversion: "cell" or "donor"

n_donors	numeric; number of donors (required if n_eff_basis = "donor")
null_fdr_threshold	numeric; FDR threshold for applicability gate (default 0.05)
on_violation	character; "error", "warn", or "pass" (default "error")
alpha	numeric; significance level for DI (default 0.05)
sided	character; "one" or "two" for DI computation (default "one")

Details

The bulknull workflow is:

1. Check applicability: bulk FDR > threshold (default 0.05)
2. If applicable, compute DDS (posterior probability of dilution)
3. Compute DI (power to detect dilution)
4. Return formatted S3 object with interpretation bands

The function preserves the full output of `dilution_score()` and `diagnosability_index()` internally for inspection if needed.

Value

An S3 object of class "bulknull" containing:

inputs	List of input parameters
applicability	List with \$applicable (logical) and \$note (character)
dds	DDS (Dilution Discordance Score) value
mu_dilution	Expected bulk z-score under dilution hypothesis
z_bulk	Observed bulk z-score
d_sc	Standardized effect size used
di	Diagnosability Index value
verdict	Interpretive string (e.g., "Ambiguous")
gate_status	Character indicating whether input passed the gate

Examples

```
# Example: DPP9 in M8 (direct d_sc input, recommended)
result <- bulknull(
  bulk_beta = -0.0297,
  bulk_se = 0.0595,
  bulk_fdr = 0.821,
  d_sc = 0.117962,
  cluster_fraction = 0.0695,
  condition_fraction = 0.631,
  alpha = 0.05,
  sided = "one"
)

print(result)
```

check_dds_applicability

Check DDS Applicability: Bulk Must Be Null

Description

Validates whether a case is applicable for DDS analysis.

DDS answers the question: "Given a bulk-NULL result and strong single-cell signal, is the null due to dilution or true absence of effect?"

This function enforces the applicability boundary:

- **Applicable:** bulk FDR > null_fdr_threshold (non-significant bulk result)
- **Not Applicable:** bulk FDR <= null_fdr_threshold (significant bulk result)

When not applicable, DDS is mathematically valid but answers the wrong question.

Usage

```
check_dds_applicability(
  bulk_fdr,
  null_fdr_threshold = 0.05,
  on_violation = c("error", "warn", "pass")
)
```

Arguments

- | | |
|--------------------|---|
| bulk_fdr | Numeric. FDR-adjusted p-value from bulk analysis. |
| null_fdr_threshold | Numeric. FDR threshold for bulk to be considered "null". Default: 0.05 (standard significance threshold). |
| on_violation | Character. One of: <ul style="list-style-type: none"> • "error": Stop with informative error message (strict mode) • "warn": Issue warning but return results (permissive mode) • "pass": Return results silently (developer/testing mode) |

Value

A list with elements:

- applicable: Logical. TRUE if bulk_fdr > null_fdr_threshold (NULL), FALSE otherwise.
- applicability_note: Character string explaining the status.
- violation_severity: Character. "none", "warning", or "error" depending on on_violation setting.
- bulk_fdr: The input FDR value.
- null_fdr_threshold: The threshold used.
- call: The matched function call.

Examples

```
# Case 1: Bulk is non-significant (FDR = 0.15) - APPLICABLE
check_dds_applicability(bulk_fdr = 0.15, null_fdr_threshold = 0.05,
                        on_violation = "warn")

# Case 2: Bulk is significant (FDR = 0.001) - NOT APPLICABLE
check_dds_applicability(bulk_fdr = 0.001, null_fdr_threshold = 0.05,
                        on_violation = "warn")
```

cohort_inflation	<i>Cohort Inflation Factor for Target Diagnosability Index</i>
------------------	--

Description

Compute the required inflation in cohort size (and thus reduction in bulk SE) to achieve a target diagnosability index. The inflation factor is independent of the pooled cohort size N and the bulk point estimate; it depends only on the observed and required effect sizes μ .

Usage

```
cohort_inflation(
  mu_observed,
  target_di = 0.8,
  alpha = 0.05,
  sided = c("one", "two")
)
```

Arguments

<code>mu_observed</code>	numeric; observed effect size (bulk z-score / $\sqrt{n_{\text{eff_bulk}}}$)
<code>target_di</code>	numeric; target diagnosability index (0 to 1, default 0.8)
<code>alpha</code>	numeric; significance level (default 0.05)
<code>sided</code>	character; "one" or "two" for one-sided or two-sided test (default "one")

Details

Identity: required inflation = $(\mu_{\text{required}} / \mu_{\text{observed}})^2$

The relationship between bulk SE and cohort size is: $SE \sim 1 / \sqrt{N}$

To reduce SE by a factor k , cohort size must increase by k^2 . Therefore, if the current μ is μ_{obs} and we need μ_{req} , the inflation is: inflation = $(\mu_{\text{req}} / \mu_{\text{obs}})^2$

This computation is independent of:

- The absolute cohort size
- The bulk point estimate (only μ_{obs} matters)
- The cluster fraction or single-cell effect size

An inflation of 50x means the cohort must grow 50 times in size. An inflation of 50 with current $N=630$ implies new $N \sim 31,500$.

Value

A numeric scalar: the required inflation factor (≥ 1.0 if target is attainable)

Examples

```
inflation <- cohort_inflation(
  mu_observed = 0.130536,
  target_di = 0.8,
  alpha = 0.05,
  sided = "one"
)

cat("Required inflation:", inflation, "\n")
cat("If current N=630, new N would be:", 630 * inflation, "\n")
```

critical_precision	<i>Critical Precision for Target Diagnosability Index</i>
--------------------	---

Description

Compute the minimum bulk measurement precision (SE) required to achieve a target diagnosability index at any cluster fraction ≤ 1 . A target DI is attainable if and only if $\text{bulk_se} \leq \text{s_critical}$, which depends only on d_sc , target_di , and α .

Usage

```
critical_precision(
  d_sc,
  bulk_se,
  target_di = 0.8,
  alpha = 0.05,
  sided = c("one", "two")
)
```

Arguments

d_sc	numeric; standardized effect size in single-cell cluster
bulk_se	numeric; observed bulk measurement standard error
target_di	numeric; target diagnosability index (0 to 1, default 0.8)
alpha	numeric; significance level (default 0.05)
sided	character; "one" or "two" for one-sided or two-sided test (default "one")

Details

The relationship is independent of the pooled cohort size N ; it depends only on the effect size standardization scale and the statistical threshold.

The condition for attainability is: $\text{bulk_se} \leq \text{d_sc} / (z_{\text{critical}} + \text{qnorm}(\text{target_di}))$

This formula is independent of N (the cohort size): it depends only on the statistical threshold and effect-size scale.

The minimum detectable effect at $f=1$ is: $\text{min_d} = \text{mu_required} * \text{bulk_se}$

For a hypothesised within-cluster effect d , the required cluster fraction is: $\text{f_required} = \text{mu_required} * \text{bulk_se} / d$

Value

A list with elements:

<code>mu_required</code>	Required effect size μ to achieve target DI
<code>s_critical</code>	Critical bulk SE; target DI attainable iff $\text{bulk_se} \leq \text{s_critical}$
<code>attainable</code>	Logical; TRUE if target DI is attainable at current bulk_se
<code>min_detectable_d</code>	Smallest within-cluster effect reachable at $f=1.0$
<code>f_required_at_d</code>	Function <code>f_required_at_d(d)</code> returning required cluster fraction
<code>call</code>	The matched function call

Examples

```
result <- critical_precision(
  d_sc = 0.117962,
  bulk_se = 0.062774,
  target_di = 0.8,
  alpha = 0.05,
  sided = "one"
)

print(result$attainable) # FALSE: bulk_se exceeds s_critical
print(result$min_detectable_d)
```

dds_bounds

DDS Bounds Over Bulk Z-Score Range

Description

Compute the attainable interval of Dilution Discordance Scores (DDS) for a given mu_dilution (effect size) as the bulk z-score z_bulk varies over a specified range.

Usage

```
dds_bounds(mu, bulk_se = NULL, z_lo = -1.96, z_hi = 1.96)
```

Arguments

<code>mu</code>	numeric; effect size μ under dilution hypothesis
<code>bulk_se</code>	numeric; bulk measurement standard error (used only for <code>min_detectable_d</code>)
<code>z_lo</code>	numeric; lower bound of bulk z-score range (default -1.96)
<code>z_hi</code>	numeric; upper bound of bulk z-score range (default 1.96)

Details

The DDS is a function of both `mu_dilution` and `z_bulk`. For fixed `mu_dilution`, extreme values of `z_bulk` produce extreme DDS values. This function returns the interval of DDS that could arise if the true bulk z-score were anywhere in the specified range (typically $[-1.96, 1.96]$ for typical bulk analyses).

Identity: $\text{DDS}(z) = \text{plogis}(\mu * z - \mu^2/2)$ where `plogis` is the logistic CDF (inverse of `logit`).

The DDS is computed as: $\text{DDS} = \text{plogis}(\mu * z - \mu^2/2)$

For fixed `mu`, as `z` varies from `z_lo` to `z_hi`, the DDS varies accordingly. The function returns the minimum and maximum attainable DDS values.

A wide interval means many DDS values are possible (uncertainty in true `z_bulk`). A narrow interval means few DDS values are possible (strong constraint on outcome).

Value

A list with elements:

<code>lower</code>	Lower bound of attainable DDS interval
<code>upper</code>	Upper bound of attainable DDS interval
<code>width</code>	Width of the interval (<code>upper - lower</code>)
<code>reaches_0.7</code>	Logical; TRUE if interval reaches $\text{DDS} \geq 0.7$
<code>reaches_0.9</code>	Logical; TRUE if interval reaches $\text{DDS} \geq 0.9$
<code>call</code>	The matched function call

Examples

```
bounds <- dds_bounds(
  mu = 0.130536,
  bulk_se = 0.062774,
  z_lo = -1.96,
  z_hi = 1.96
)

cat("DDS range:", bounds$lower, "to", bounds$upper, "\n")
cat("Interval width:", bounds$width, "\n")
cat("Reaches strong evidence (0.7)?", bounds$reaches_0.7, "\n")
```

diagnosability_index *Diagnosability Index (DI)*

Description

Compute statistical power to detect dilution under a one-sided or two-sided test.

Usage

```
diagnosability_index(mu_dilution, alpha = 0.05, sided = c("one", "two"))
```

Arguments

mu_dilution	Numeric. Expected signal under dilution: $\mu = (\text{cluster_frac} * \text{sc_zscore}) / (\text{bulk_se} * \text{sqrt}(\text{n_eff}))$.
alpha	Numeric. Significance level (default 0.05).
sided	Character. "one" for one-sided test (default, recommended for directional alternatives) or "two" for two-sided.

Details

DI estimates the probability of correctly detecting dilution if it truly occurred at the specified effect size (mu_dilution).

One-sided (default, recommended):

- Critical value: $z = \text{qnorm}(1 - \alpha) = 1.644854$ for $\alpha=0.05$
- Rationale: Dilution alternative is directional by construction ($\text{sign}(\mu_dilution) = \text{sign}(\text{sc_zscore})$ since $\text{cluster_frac} > 0$)
- Formula: $DI = \text{pnorm}(\mu - 1.644854)$

Two-sided (for reference):

- Critical value: $z = \text{qnorm}(1 - \alpha/2) = 1.959964$ for $\alpha=0.05$
- Formula: $DI = \text{pnorm}(\mu - 1.959964)$

Power categories:

- $DI < 0.3$: Low power (undetectable at this effect size)
- $DI 0.3-0.8$: Moderate power (detection possible but unreliable)
- $DI > 0.8$: High power (reliable detection)

Value

List containing:

- di: Diagnosability Index (power), in range 0 to 1
- critical_value: Critical z-value used
- alpha: Significance level
- sided: Test type ("one" or "two")
- formula: Human-readable formula string

Examples

```
# DPP9/M8 k=5 tissue-only
diagnosability_index(mu_dilution = 0.130536, alpha = 0.05, sided = "one")
# Returns: di = 0.0650 (very low power; dilution undetectable)
```

dilution_scan

*Vectorized Dilution Scan Across Multiple Genes***Description**

Applies the bulknull framework to multiple genes in a single call, returning a summary table with DDS, DI, and verdict for each gene. Skips bulk-significant genes ($FDR \leq \text{threshold}$) and genes with missing single-cell entries.

Usage

```
dilution_scan(
  bulk_deg,
  d_sc_values = NULL,
  sc_zscores = NULL,
  cluster_fraction,
  condition_fraction,
  n_cluster = NULL,
  n_eff_basis = c("cell", "donor"),
  n_donors = NULL,
  null_fdr_threshold = 0.05,
  alpha = 0.05,
  sided = c("one", "two")
)
```

Arguments

bulk_deg	data frame; differential expression results with columns gene (character), beta or log2fc (numeric effect size), se or se_log2fc (numeric standard error), and fdr or padj (numeric FDR). Additional columns are ignored.
d_sc_values	named numeric vector; standardized effect sizes keyed by gene (primary input)
sc_zscores	named numeric vector; z-scores keyed by gene (backward compatibility). Genes not in either vector are skipped with a note.
cluster_fraction	numeric; fraction of total cells in target cluster (constant across all genes).
condition_fraction	numeric; fraction of cells in case condition (constant across all genes).
n_cluster	numeric; number of cells in target cluster (constant across all genes). Required if sc_zscores supplied.
n_eff_basis	character; "cell" or "donor" basis for z-score conversion.
n_donors	numeric; number of donors (required if n_eff_basis = "donor").
null_fdr_threshold	numeric; FDR threshold for applicability gate (default 0.05). Genes with $FDR \leq \text{threshold}$ are marked "FAILED".
alpha	numeric; significance level for DI (default 0.05).
sided	character; "one" or "two" for one-sided or two-sided test (default "one").

Details

The function iterates over rows of `bulk_deg`, skipping:

- Genes with bulk FDR $\leq$ `null_fdr_threshold` (gate fails)
- Genes without an entry in `sc_zscores`
- Genes with zero variance in the single-cell data (`sc_zscore` = 0)

For applicable genes, it calls `bulknull()` internally and extracts key results. Returns all processed genes in output, with NA or "SKIPPED" for missing data.

Value

A data frame with columns:

<code>gene</code>	Gene name from <code>bulk_deg</code>
<code>bulk_beta</code>	Effect size from <code>bulk_deg</code>
<code>bulk_se</code>	Standard error from <code>bulk_deg</code>
<code>bulk_fdr</code>	FDR from <code>bulk_deg</code>
<code>sc_zscore</code>	Single-cell z-score (NA if not found)
<code>mu_dilution</code>	Expected bulk z-score under dilution hypothesis
<code>dds</code>	Dilution Discordance Score
<code>di</code>	Diagnosability Index
<code>dds_interpretation</code>	Categorical interpretation of DDS
<code>di_interpretation</code>	Categorical interpretation of DI
<code>verdict</code>	Final interpretive verdict
<code>gate_status</code>	PASSED or FAILED (applicability gate status)

Examples

```
# Example: scan a small DEG table with mock z-scores
bulk_deg <- data.frame(
  gene = c("GENE1", "GENE2", "GENE3"),
  beta = c(-0.084, 0.020, 0.150),
  se = c(0.063, 0.050, 0.100),
  fdr = c(0.450, 0.05, 0.001)
)
sc_zscores <- c(GENE1 = 2.077, GENE2 = 1.5, GENE3 = 0.5)

result <- dilution_scan(
  bulk_deg = bulk_deg,
  sc_zscores = sc_zscores,
  cluster_fraction = 0.0695,
  condition_fraction = 0.631,
  n_cluster = 1332
)
print(result)
```

dilution_score	<i>Dilution Discordance Score (DDS)</i>
----------------	---

Description

Compute the Bayesian posterior probability that an observed bulk-level effect reflects true dilution of a single-cell signal.

Usage

```
dilution_score(
  bulk_beta,
  bulk_se,
  bulk_fdr,
  d_sc = NULL,
  sc_zscore = NULL,
  cluster_fraction,
  condition_fraction,
  n_cluster = NULL,
  n_eff_basis = c("cell", "donor"),
  n_donors = NULL,
  null_fdr_threshold = 0.05,
  on_violation = c("error", "warn", "pass")
)
```

Arguments

bulk_beta	Numeric. Observed bulk effect size (Hedges' g or log2-fold-change).
bulk_se	Numeric. Standard error of bulk effect size.
bulk_fdr	Numeric. FDR-adjusted p-value from bulk analysis (or raw p-value). Used for validation only; not directly in likelihood.
d_sc	Numeric. Standardized effect size in the single-cell cluster (primary input). When supplied, sc_zscore conversion is skipped.
sc_zscore	Numeric. Single-cell z-score (backward compatibility). Ignored if d_sc supplied. Must provide either d_sc or sc_zscore, not both unless testing.
cluster_fraction	Numeric. Fraction of total tissue represented by the responsible subcluster ($0 < f \leq 1$).
condition_fraction	Numeric. Fraction of subcluster cells from the condition of interest (e.g., IPF, $0 \leq p \leq 1$).
n_cluster	Integer. Number of cells in the subcluster. Required if sc_zscore supplied.
n_eff_basis	Character. Basis for sc_zscore conversion: "cell" or "donor". Default "cell". Ignored if d_sc supplied.
n_donors	Integer. Number of donors. Required if n_eff_basis = "donor".
null_fdr_threshold	Numeric. FDR threshold for bulk to be considered "null". Default: 0.05. Only used if bulk_fdr is provided.
on_violation	Character. One of "error", "warn", "pass". Controls behavior when bulk FDR is significant. Default: "warn".

Value

A list with elements:

- dds_score: Posterior probability of dilution (0 to 1).
- z_bulk: Bulk z-score (beta / SE).
- mu_dilution: Expected z under dilution model.
- d_sc: Standardized effect size used (supplied or converted).
- log_lik_h1: Log-likelihood under dilution model.
- log_lik_h0: Log-likelihood under null model.
- summary: Plain-text interpretation.
- applicable: Logical. TRUE if case is applicable (bulk is null), FALSE if not.
- applicability_note: Character string describing applicability status.
- call: The matched function call.

Examples

```
# Direct d_sc input (recommended)
dilution_score(
  bulk_beta = -0.0297,
  bulk_se = 0.0595,
  bulk_fdr = 0.821,
  d_sc = 0.117962,
  cluster_fraction = 0.0695,
  condition_fraction = 0.631
)

# Backward compatible: sc_zscore with cell-level basis
dilution_score(
  bulk_beta = -0.0297,
  bulk_se = 0.0595,
  bulk_fdr = 0.821,
  sc_zscore = 2.077,
  cluster_fraction = 0.0695,
  condition_fraction = 0.631,
  n_cluster = 1332,
  n_eff_basis = "cell"
)
```

dpp9_m8

DPP9/M8 Example Dataset

Description

A list containing measurements for DPP9 (diphosphonucleotidase 9) expression in the M8 (inflammatory macrophages) subcluster from IPF lung tissue.

Usage

```
data(dpp9_m8)
```

Format

A list with 12 elements:

bulk_beta Numeric. Bulk effect size (-0.0297)
bulk_se Numeric. Standard error of bulk effect (0.0595)
bulk_fdr Numeric. FDR-adjusted p-value from meta-analysis (0.821)
sc_zscore Numeric. Single-cell module z-score (2.077)
cluster_fraction Numeric. Fraction of tissue in M8 (0.0695)
condition_fraction Numeric. Fraction of M8 from IPF (0.631)
n_cluster Integer. Number of cells in M8 (1332)
gene Character. Gene name ("DPP9")
cluster Character. Cluster identifier ("M8")
cluster_name Character. Cluster description ("Inflammatory Macrophages")
cohort Character. Disease cohort ("IPF")
description Character. Dataset description

Details

This is a real data example from the bulknull application: a bulk-level null result (non-significant effect) paired with strong single-cell signal in a small subcluster. Used throughout vignettes and examples.

The DPP9/M8 case exemplifies the framework's utility: bulk RNA-seq from IPF lung tissue shows no significant DPP9 effect (FDR = 0.821), but single-cell analysis reveals strong inflammasome-module enrichment in M8. The DDS computes whether the bulk null reflects true absence or dilution of this M8 signal by the dominant other cell types.

Result: DDS ~ 0.48 (ambiguous), suggesting both null and dilution remain plausible. The Diagnosticity Index (DI ~ 0.065) reveals that the study design is fundamentally underpowered for dilution detection due to the small M8 fraction (6.95% of lung), independent of effect size.

Examples

```
data(dpp9_m8)
str(dpp9_m8)

# Compute Dilution Discordance Score
result <- dilution_score(
  bulk_beta = dpp9_m8$bulk_beta,
  bulk_se = dpp9_m8$bulk_se,
  bulk_fdr = dpp9_m8$bulk_fdr,
  sc_zscore = dpp9_m8$sc_zscore,
  cluster_fraction = dpp9_m8$cluster_fraction,
  condition_fraction = dpp9_m8$condition_fraction,
  n_cluster = dpp9_m8$n_cluster
)
print(result$dds_score)
```

plot.bulknull

*Plot DI Landscape for Study Design***Description**

Visualizes the Diagnosability Index as a function of cluster fraction and bulk measurement precision (SE), marking the current study's position. Base graphics only.

Usage

```
## S3 method for class 'bulknull'
plot(
  x,
  sc_zscore = NULL,
  condition_fraction = NULL,
  n_cluster = NULL,
  alpha = 0.05,
  sided = c("one", "two"),
  cluster_fraction_range = c(0.001, 1),
  se_range = c(0.001, 0.5),
  grid_points = 50,
  ...
)
```

Arguments

x	A bulknull object
sc_zscore	numeric; single-cell z-score (same as input to bulknull)
condition_fraction	numeric; case fraction in cohort
n_cluster	numeric; cells in target cluster
alpha	numeric; significance level (default 0.05)
sided	character; "one" or "two" (default "one")
cluster_fraction_range	numeric vector of length 2; range for y-axis
se_range	numeric vector of length 2; range for x-axis (log scale)
grid_points	numeric; number of grid points per dimension (default 50)
...	Additional arguments (ignored)

Details

The landscape shows DI as a 2D function of cluster_fraction and bulk_se. Higher DI (>0.8) indicates good power; lower DI (<0.3) indicates underpowered. Contours mark DI = 0.3 and DI = 0.8 boundaries. The study is marked with a red dot at (bulk_se, cluster_fraction) from the input bulknull object.

print.bulknull	<i>Print method for bulknull objects</i>
----------------	--

Description

Displays a formatted verdict block showing inputs, computed statistics, interpretations, and the final verdict.

Usage

```
## S3 method for class 'bulknull'  
print(x, ...)
```

Arguments

x	A bulknull object
...	Additional arguments (ignored)

print.critical_precision	<i>Print method for critical_precision objects</i>
--------------------------	--

Description

Print method for critical_precision objects

Usage

```
## S3 method for class 'critical_precision'  
print(x, ...)
```

Arguments

x	A critical_precision object
...	Additional arguments (ignored)

print.dds_bounds	<i>Print method for dds_bounds objects</i>
------------------	--

Description

Print method for dds_bounds objects

Usage

```
## S3 method for class 'dds_bounds'
print(x, ...)
```

Arguments

x	A dds_bounds object
...	Additional arguments (ignored)

print.required_design	<i>Print method for required_design objects</i>
-----------------------	---

Description

Print method for required_design objects

Usage

```
## S3 method for class 'required_design'
print(x, ...)
```

Arguments

x	A required_design object
...	Additional arguments (ignored)

required_design	<i>Required Study Design to Achieve Target Diagnosability Index</i>
-----------------	---

Description

Inverts the diagnosability index formula to find the study parameters needed to achieve a target power (DI). Can solve for either cluster_fraction (at fixed bulk_se) or bulk_se (at fixed cluster_fraction).

Usage

```
required_design(
  target_di,
  d_sc = NULL,
  sc_zscore = NULL,
  condition_fraction,
  n_cluster = NULL,
  n_eff_basis = c("cell", "donor"),
  n_donors = NULL,
  cluster_fraction,
  bulk_se,
  alpha = 0.05,
  sided = c("one", "two")
)
```

Arguments

target_di	numeric; target diagnosability index (0 to 1)
d_sc	numeric; standardized effect size in single-cell cluster (primary input)
sc_zscore	numeric; single-cell z-score (backward compatibility)
condition_fraction	numeric; fraction of cells in case condition (0 to 1)
n_cluster	numeric; number of cells in target cluster (required if sc_zscore supplied)
n_eff_basis	character; basis for conversion: "cell" or "donor"
n_donors	numeric; number of donors (required if n_eff_basis = "donor")
cluster_fraction	numeric; fraction of total cells in target cluster
bulk_se	numeric; standard error of bulk effect estimate
alpha	numeric; significance level (default 0.05)
sided	character; "one" or "two" for one-sided or two-sided test (default "one")

Details

The diagnosability index is defined as: $DI = \text{Normal_CDF}(\mu - z_{\text{critical}})$ where Normal_CDF is the standard normal CDF and z_{critical} depends on alpha and sided.

This function inverts the formula in two ways:

1. Solves for cluster_fraction given bulk_se, sc_zscore, and condition_fraction
2. Solves for bulk_se given cluster_fraction, sc_zscore, and condition_fraction

If the required cluster_fraction exceeds 1 (unattainable), the function returns NA and includes a warning message.

Value

A list with class "required_design" containing:

target_di	Requested diagnosability index
achieved_di_at_fixed_se	DI achieved if solving for cluster_fraction

```

required_cluster_fraction      Cluster fraction needed to reach target DI (at fixed bulk_se)
required_bulk_se               Bulk SE needed to reach target DI (at fixed cluster_fraction)
achieved_di_at_fixed_cf       DI achieved if solving for bulk_se
scenario_fixed_se              Message describing the fixed-SE scenario
scenario_fixed_cf              Message describing the fixed-CF scenario
warnings                       Character vector of warnings if targets are unattainable

```

Examples

```

# Example: k=5 tissue-only case study
# Current DI is only 0.065; what would we need for DI = 0.3?

design <- required_design(
  target_di = 0.3,
  sc_zscore = 2.077409,
  condition_fraction = 0.631,
  n_cluster = 1332,
  cluster_fraction = 1332 / 19175,
  bulk_se = 0.062774,
  alpha = 0.05,
  sided = "one"
)

print(design)

```

summary.bulknnull

Summary method for bulknnull objects

Description

Displays a concise one-line summary of the diagnosis.

Usage

```

## S3 method for class 'bulknnull'
summary(object, ...)

```

Arguments

```

object      A bulknnull object
...         Additional arguments (ignored)

```